EMPLOYEE SALARY & HEADCOUNT

Data Analysis Report | Prepared by Hassan Jumaa (Riq)

PYTHON · MYSQL · SQL · POWER BI

PROJECT OVERVIEW

Project Overview

This report summarises an end-to-end employee data analysis project: cleaning a raw HR dataset in Python (Pandas), loading it into MySQL, querying it with SQL, and visualising the results in an interactive Power BI dashboard. The dataset covers **10 employees** across **5 Kenyan cities** (Eldoret, Kisumu, Mombasa, Nairobi, Nakuru).

TOOLS & WORKFLOW

Tools & Workflow

Stage	Tool	Purpose
Data Cleaning	Python (Pandas, NumPy)	Remove duplicates, handle nulls, fix invalid ages, repair salary digit slips
Storage	MySQL	Persist the cleaned dataset for querying and reporting
Querying	SQL	Aggregate and rank employees by salary, city and hire date
Visualization	Power BI	Interactive dashboard with KPI cards, charts and slicers
Reporting	ReportLab	Generate this PDF programmatically from the data

DATA CLEANING SUMMARY

Data Cleaning Summary

The raw export contained **8 data defects** (plus one date-format standardisation applied to every row). Each was resolved deliberately and documented, never silently.

Issue Found	Affected Record	Resolution Applied
Duplicate employee records	ID 1, ID 4 — Alice	Dropped the duplicate. Keeping it would double-count headcount and payroll.
Missing employee name	EmployeeID 9 — name blank	Filled with the placeholder **'Jamal'**. Names are not statistically recoverable; this is an **assumption** and carries no analytical weight.
Missing age values	EmployeeID 2 — Bob age empty	Imputed with the median age (30), which is robust to outliers.
Invalid negative age	EmployeeID 7 — Faith age = -5	Treated as missing, then filled with the same median age (30) and rounded to whole years.
Missing salary value	EmployeeID 3 — Charlie salary blank	Imputed with the median salary (KES 61,000). The mean is unusable for this purpose because the digit slips in the same column would dominate it.
Missing city value	EmployeeID 5 — Diana city empty	Filled with **Kisumu** as a **disclosed assumption, not a recovery**. The export holds no evidence for any particular city, and every Kisumu figure in this report moves if the choice is wrong. See Limitations.
Salary digit slip (one digit too many)	EmployeeID 8 — George salary KES 900,000	Repaired to KES 90,000 by shifting one power of ten (divided by 10) — which lands back inside the plausible band. The repair is reported in the run output, never silent.

Issue Found	Affected Record	Resolution Applied
Salary digit slip (one digit too few)	EmployeeID 11 — Kirauti salary KES 7,500	Repaired to KES 75,000 by shifting one power of ten (multiplied by 10) — which lands back inside the plausible band. The repair is reported in the run output, never silent.
Date format standardised	All records	Parsed as day-first dates and standardised to a single format, so the dates sort and group correctly by month.

Row IDs refer to the raw export — the cleaned file renumbers rows, so raw ID 5 (Diana) is ID 4 there.

KEY METRICS

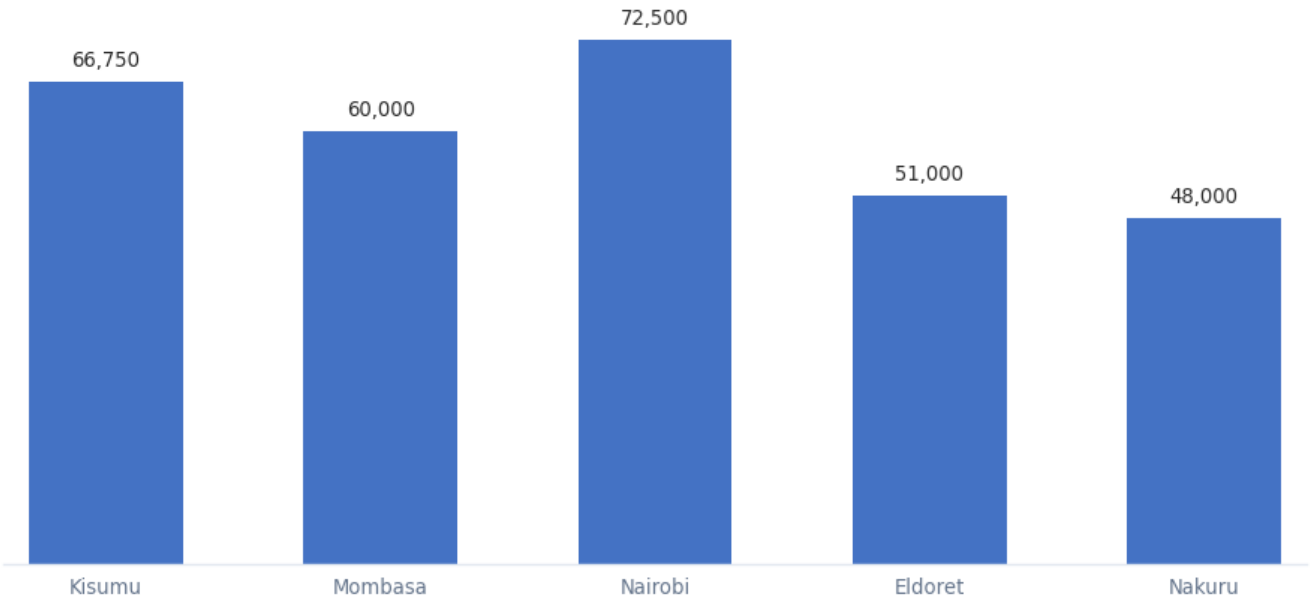
Key Metrics

10	KES 631,000	KES 63,100	30.5
Total Employees	Total Salary Expense	Average Salary	Average Age

Median salary is **KES 61,000** against a mean of **KES 63,100**. The mean sits above the median, so the distribution is **right-skewed**: a small number of high earners pull the average above what a typical employee earns. Range is **KES 42,000** (KES 48,000 → KES 90,000).

SALARY DISTRIBUTION BY CITY

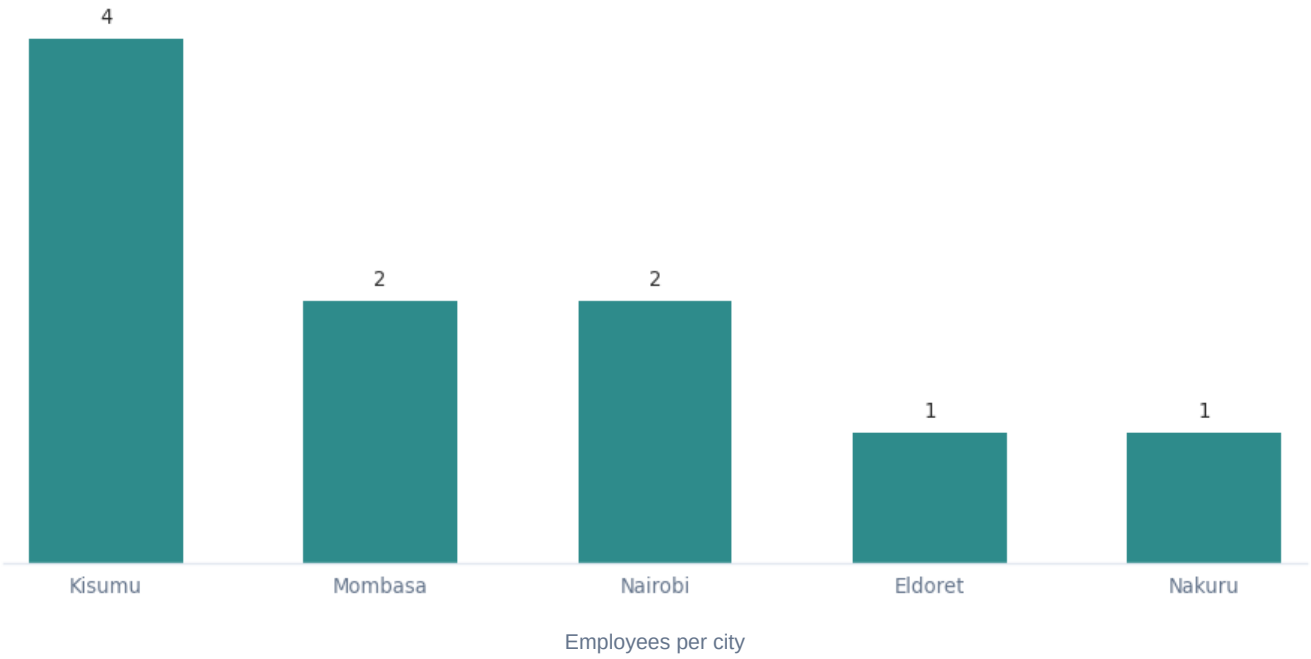
Salary distribution by city



Average salary by city — KES

HEADCOUNT BY CITY

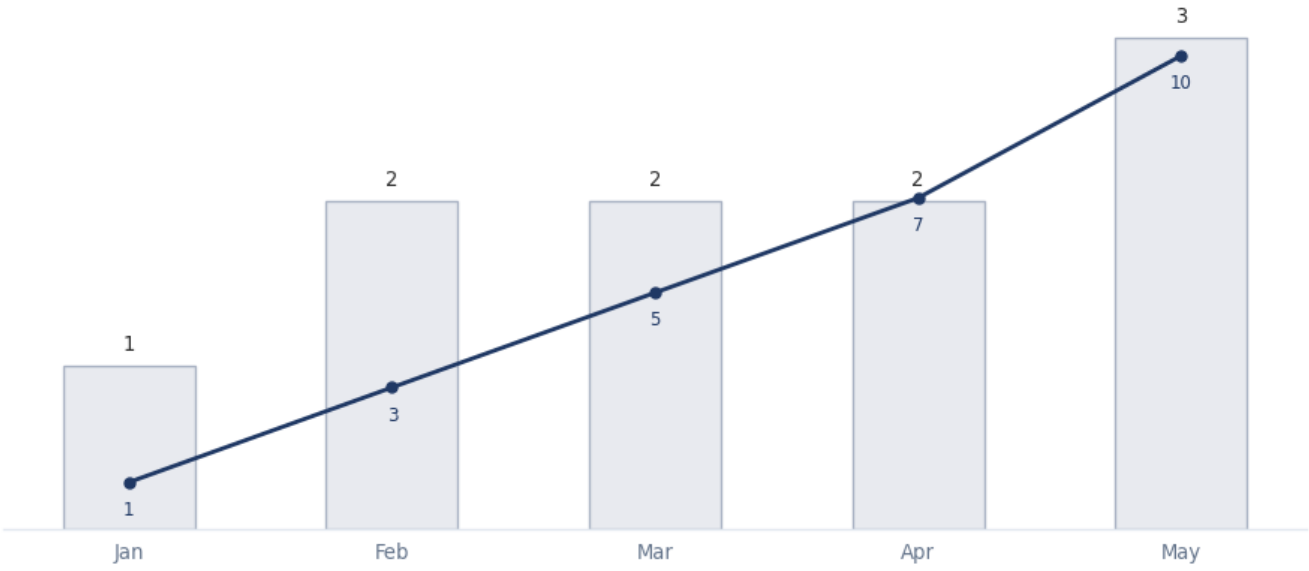
Headcount by city



Employees per city

HIRING TREND BY MONTH

Hiring trend by month



New hires per month, 2025

Headcount grew from **1 hire in Jan** to **10 employees** by May, so the monthly payroll bill ramped **KES 55,000** (Jan) → **KES 178,000** (Feb) → **KES 296,000** (Mar) → **KES 437,000** (Apr) → **KES 631,000** (May). The KES 631,000 figure is the **current** monthly bill at full headcount — not a constant cost across the period.

EMPLOYEE SALARY RANKING

Employee Salary Ranking

#	Name	City	Salary (KES)	vs. Average
1	**George**	Nairobi	90,000	+42.6%
2	**Kirauti**	Kisumu	75,000	+18.9%
3	**Diana**	Kisumu	70,000	+10.9%
4	**Bob**	Mombasa	62,000	-1.7%
5	**Charlie**	Kisumu	61,000	-3.3%
6	**Jamal**	Kisumu	61,000	-3.3%
7	**Hassan**	Mombasa	58,000	-8.1%
8	**Alice**	Nairobi	55,000	-12.8%
9	**Faith**	Eldoret	51,000	-19.2%
10	**Evan**	Nakuru	48,000	-23.9%

KEY INSIGHTS

Key Insights

01 **City salary disparity.** Nairobi leads with an average of KES 72,500 (+14.9% vs. the company mean), while Nakuru is lowest at KES 48,000 (-23.9%). Nairobi sits 51% above Nakuru.

02 **Headcount distribution.** Kisumu 4, Nairobi 2, Mombasa 2, Nakuru 1 and Eldoret 1 employees across the 5 cities.

03 **Age drives pay.** Pearson correlation between age and salary is **r = 0.765** ($r^2 = 0.59$) — age alone explains roughly 59% of salary variance. Alice, the youngest at 25, earns KES 55,000; George, the oldest at 40, earns KES 90,000 — a **1.64×** gap.

04 **One earner the average cannot absorb.** George's KES 90,000 is the highest salary in the set. Across all 10 salaries the Tukey upper fence is KES 98,750 (Q1 = 55,000, Q3 = 72,500, IQR = 17,500), and no employee is flagged against it. Measured against the other 9 salaries the fence falls to KES 85,500, and KES 90,000 still sits above it — the one figure the average cannot absorb.

> **THE FINDING THAT MATTERS: CLEANING CHANGED THE CONCLUSION**
>
> Computed on the raw export, average salary reads **KES 136,750**. Correctly cleaned it is **KES 63,100** — an inflation of **117%**, caused by 2 digit slips in the salary column and one duplicated row. Left untreated, this dataset would have produced a false narrative about what the company pays. The cleaning step *is* the analysis.

LIMITATIONS & HONEST SCOPE

Limitations & Honest Scope

Two values in this dataset are **assumptions made during cleaning, not facts recovered from the source**. They are stated plainly so the numbers above can be trusted correctly.

Value	Status	Why it is not recoverable	Impact
Diana's City → Kisumu	Assumption	Location cannot be derived from age, salary or join date. The export holds no evidence for any particular city, so the value is a disclosed placeholder rather than a recovered fact.	Kisumu averages **KES 66,750** with the assumption and **KES 65,667** without it; its headcount reads **4 of 10** with it and **3** without. Which city has the most people does not change either way.
ID 9's Name → 'Jamal'	Placeholder	Names carry no analytical weight — no calculation touches this field.	None.

If this were a real stakeholder deliverable: both fields would be left as *Unknown*, Diana excluded from the city comparison, and the location view reported at 9 of 10 employees. Total salary, mean, median, the inflation finding and the distribution analysis are identical under either choice — only the Kisumu city average and headcount move.

> **RECOMMENDATION**
>
> 1. Introduce age/experience-based pay bands — an r^2 of 0.59 says seniority is already driving pay, just informally. 2. Review the top earner against market benchmarks before treating Nairobi's average as a real signal. 3. Add validation at source: a primary-key constraint, an age range check and a salary sanity cap would have caught all 8 defects before they reached a report.